

1. Retrieval is the safety layer

An AI answer is only as reliable as the evidence it retrieves

- FDA evidence is precise: **product, strength, route, dosage form, and regulatory status** all matter.
- Retrieving the wrong cohort can still produce a fluent, convincing answer.
- Generation cannot recover evidence that retrieval missed—or detect evidence that retrieval mixed incorrectly.

Better retrieval is more important than better wording.

Goal: make retrieval observable, testable, and capable of improving from failure.

2. FDA retrieval is a graph problem

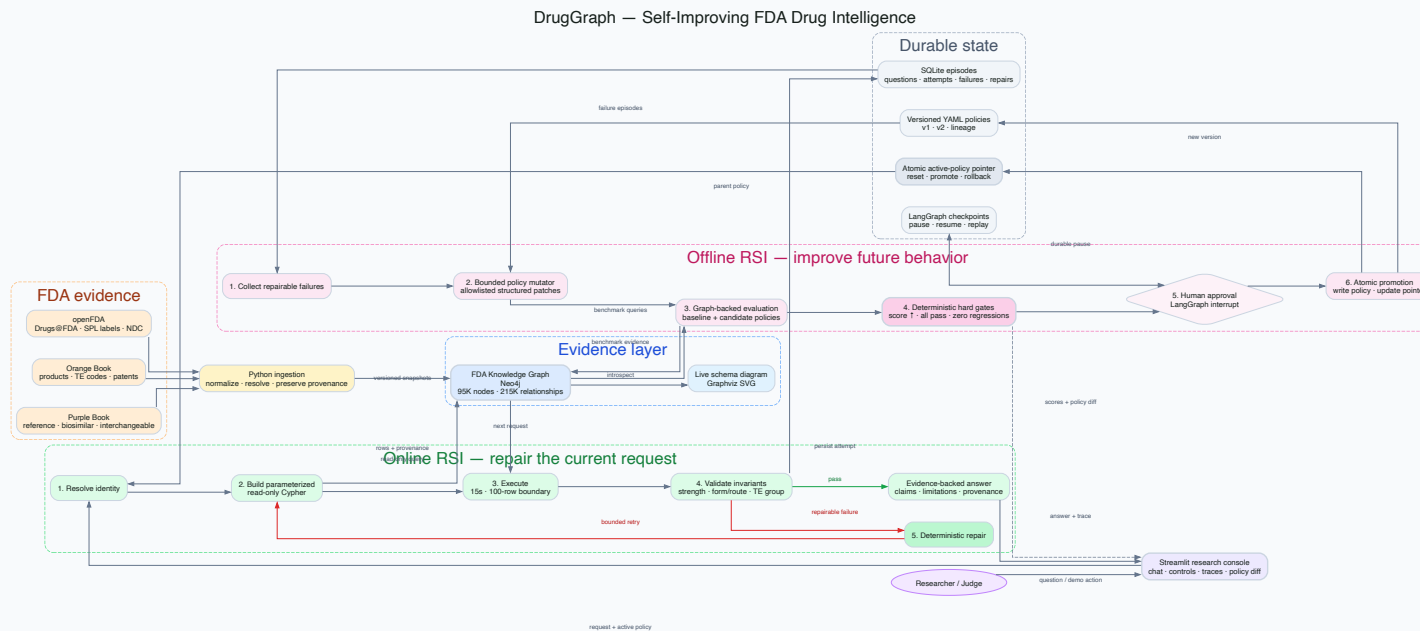
Source	Evidence
Orange Book	Products, applications, RLD/RS status, TE codes, patents
Purple Book	Reference biologics, biosimilars, interchangeable products
openFDA	Drugs@FDA, SPL labels, and NDC listings

The graph connects

Drug names → products → applications → regulatory evidence → labels

- Approximately **95K nodes** and **215K relationships**
- Provenance remains attached to every claim
- State substitution law remains a separate, versioned source

3. DrugGraph system design



- **Neo4j:** FDA evidence and provenance
- **LangGraph:** online repair and offline optimization
- **SQLite + YAML:** durable episodes, checkpoints, policy lineage, and rollback

4. Self-improving retrieval—not self-modifying code

Detect failure → Repair request → Record episode
↓
Generate bounded policies → Run graph-backed evals
↓
Hard safety gates → Human approval → Promote → Replay

Demonstrated policy evolution

	Policy v1	Promoted v2
Evaluation score	37.5%	100%
Lipitor query attempts	2	1
Adalimumab capability	Blocked	Enabled after Purple Book gate

Promotion rule: higher score + all hard checks + zero regressions + approval. Every mutation is allowlisted and shown as a YAML diff.

5. What worked—and what comes next

What worked

- Deterministic FDA invariants caught errors that an LLM judge might miss.
- Graph-backed evaluation made policy improvement measurable.
- Approval, versioning, reset, and rollback made the loop demoable and auditable.

Next

- Add an **LLM judge** for usefulness and clarity—never as the sole promotion gate.
- Add an LLM proposer that emits only structured, allowlisted policy patches.
- Evaluate on frozen FDA graph snapshots and shadow production traffic.
- Expand RSI to label monitoring, biologic navigation, and sourced state-law rules.

DrugGraph turns retrieval failures into tested, durable improvements.